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SUPERSEDING
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MILITARY SPECIFICATION

COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a one-piece protective coverall made of polyester and cotton twill cloth.

1.2 Classification. The coveralls shall be of one type and in the following sizes (see 6.2):

Small - Medium - (Up to 43)
Large - X-Large - (44 to 48)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATION

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer-in-Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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FEDERAL

V-T-276 - Thread, Cotton
JJ-S-746 - Stockinet, Surgical
JJ-W-155 - Webbing, Textile (Cotton, Elastic)
DDD-L-20 - Label; for Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-G-16491 - Grommet, Metallic
MIL-B-17757 - Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic
MIL-C-21851 - Cloth, Twill, Polyester and Cotton

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

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(Application for copies should be addressed to American Trucking Associations, ATTN: Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials. (see 6.4).

3.3.1 Basic material. The basic material shall be a polyester and cotton twill cloth, shade yellow 3511 conforming to MIL-C-21851. When no standard sample is available, the colorfastness to light shall be 20 standard fading hours.

3.3.2 Thread, cotton. The cotton thread used for seaming and stitching shall be ticket nos. 30 and 50, 3 ply, Type 1A3 of V-T-276. The thread shall be vat dyed to approximate yellow AS, Cable No. 66059.

3.3.3 Webbing, elastic. The webbing shall be cotton, elastic, unbleached or bleached white conforming to the types and classes of JJ-W-155 as follows or a woven or braided elastic webbing of equivalent grade may be used.

Type	Class	Width	Use
I	10	1-1/2 inch	Back waist take-up
I	7	1 inch	Foot stirrups
II	1	5/16 inch	Hem at leg bottom

3.3.4 Wristlets, knitted, cotton. The knitted cotton wristlets shall be bleached or unbleached, 3 inches wide, conforming to type I of JJ-S-746.

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3.3.5 Braid, nylon. The drawcord for the top of the coveralls shall be braided of multifilament nylon, and shall conform to the requirements specified in Table I when tested as specified in 4.4.1. The finished braid shall be not less than 3/16 inch in width and the color shall be white.

Table I - Physical characteristics

Characteristic	Requirements
Yarn denier	840 denier
Number of carriers	16
Ends per carrier	3
Picks per inch	11 (min.)

3.3.6 Grommets and washers. The grommets and washers shall conform to type I, class 2, size 1-J of MIL-G-16491.

3.3.7 Fastener tapes. The hook and pile fastener tapes shall conform to type II, class 1 of MIL-F-21840. White 3054 tape fasteners shall be used in "small-medium" size coveralls and Red 3852 tape fasteners in "large-X-large" size coveralls to provide ready size identification. Tapes shall be as follows:

Width	Tape	Use
5/8 inch	Hook	Pocket flap
1 inch	Pile	Pocket
2 inch	Pile	Center back

3.3.8 Combination label. Each coverall shall have a combination identification, size and instruction label conforming to type VI, class 14 of DDD-L-20. The printing shall show colorfastness to laundering. The inscription shall be as follows:

COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE
FIBER CONTENT: 65% Polyester and 35% Cotton
CONTRACT NO.: DLA 100-00-0-0000 (Example)
STOCK NO.: 8415-00-000-0000 (Example)
SIZE: Small-Medium, up to size 43 (Example)
NAME OF SUPPLIER:

LAUNDER IN ACCORDANCE WITH PRESCRIBED REQUIREMENTS

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3.3.9 Radiation Symbol. The marking for the radiation symbol shall be in accordance with type IV of DDD-L-20. The radiation symbol shall be 3-3/4 inches in diameter conforming to the design and dimensions as shown in Figure 2.

3.4 Design. The coveralls, as shown in Figure 1, shall be one piece having full length sleeves with knitted wristlets, a two-section left breast pocket with abutted flaps, and a channeled nylon drawcord at neck closure. It shall incorporate elastic webbing for back waist take-up, for drawing in around leg bottoms, and foot stirrup straps. A black radiation symbol shall be stamped on upper left back of coverall.

3.4.1 Figures. The figures are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns will be furnished by the Government to the contractor for use in cutting working patterns (see 6.3). The working patterns shall be identical to the standard patterns. Neither the Government patterns nor the working patterns shall be altered in any way. The patterns provide a seam allowance of 1/2 inch for all regular double lapped, double stitched seams.

3.5.1 Pattern parts. The component parts of the coveralls shall be cut from material as specified and in accordance with the following pattern parts:

Material	Nomenclature of pattern parts	Cut parts
Cloth, polyester/ cotton twill	Front - Back	2
	Front	2 1/
	Back	2 1/
	Sleeve, front	2
	Sleeve, back	2
	Pocket, double (dosimeter and film badge)	1
	Pocket flap (dosimeter)	1
	Pocket flap (film badge)	1

1/ If the width of the material does not allow for economical cutting, the size large-X Large patterns, which provide for a four-piece body, may be used.

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3.6 Stitches, seams and stitchings. The stitches, seams, and stitching types specified in Table II shall conform to FED-STD-751. Whenever two or more methods for seam or stitch type are given for the same operation, any one may be used. Unless otherwise specified, all double needle seaming shall be 1/4 inch gauge with the row of stitching next to the folded edge not more than 1/16 inch from edge. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the coveralls.

3.6.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or 401 stitch type thread breaks may be repaired by using 301 stitch type. When stitch type 401 is permitted, as an option, it shall not be used unless the ends are caught in other seams or stitching.

3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with the operation requirements specified in Table II. The contractor is not required to follow the exact sequence of operations listed provided that the finished coveralls are identical to those produced by following the sequence of operations as listed in Table II.

3.7.1 Shade and size marking. The component parts of the coveralls shall be marked, ticketed or bundled to insure a uniform shade and size throughout the garment. Any method may be used except:

1. Corrosive metal devices.
2. Sew-on shade tickets.
3. Adhesive type tickets which discolor or adhere to the material upon removal of tickets.

NOTE: The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the garment and wherever possible, is covered by the seam allowance.

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Table II - Construction of coverall

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. The coverall shall be cut in strict accordance with patterns furnished, which shall show size, placement of component parts, directional lines for cutting, and notches for proper assembly of all parts. Directional lines shall be placed in the warp direction of the material and may vary from the warp direction by not more than 1-1/2 inches. Measurements shall be taken at the top and bottom of the directional line on patterns to the selvage edge of the fabric on the even side of the lay. The difference between the two measurements shall not exceed 1-1/2 inches. b. All component parts shall be cut from one piece of material. The filling flush side shall be the face and shall finish as the exposed surface of the coveralls.					
2. *	Replacement of defective components During the spreading, cutting and manufacturing process, components parts having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Marking All parts shall be marked, bundled or ticketed to insure proper assembly throughout the coverall (see 3.7.1).					

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Table II - Construction of coverall (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Pocket (dosimeter and film badge)					
	a. Hem top of pocket with edge turned under 1/4 inch. Hem shall measure 5/8 (± 1/8) inch wide.	301	EFb-1	10-12	30	30
	b. Position 1 inch wide pile fastener tape 1/4 inch from folded top edge of pocket across full width of finished pocket. Stitch on all four sides 1/16 to 1/8 inch from edge.	301	LSbj-1	10-12	30	30
	c. Double stitch 1/4 inch gauge the sides and bottom edge of pocket to left front, according to notches, with edges turned under.	301	LSd-2	10-12	30	30
	d. Double stitch 1/4 inch gauge on pocket to form separate dosimeter and film badge sections. All stitching shall cross pile fastener tape.	301	SSv-2	10-12	30	30
	e. The inside measurements for the finished pocket sections shall be as follows: Dosimeter 2 (± 1/8) inches wide and 4-3/4 (± 1/4) inches deep. Film badge 2-5/8 (± 1/8) inches wide and 3-1/2 (± 1/4) inches deep.					
5.	Pocket flaps					
	a. Fold flap, face to face, and stitch sides 1/4 inch from edge. Turn to finished position and work out point.	301	SSa-1	10-12	30	30

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Table II - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Pocket flaps (cont'd)					
	b. Position 5/8 inch hook fastener tape to underside of pocket flaps full width of finished flaps, and stitch 1/16 to 1/8 inch from edge on all four sides.	301	LSbj-1	10-12	30	30
	c. Stitch underside of pocket flap to left front according to notches 1/8 to 3/16 inch from raw edge. Turn to first position and stitch 3/16 to 1/4 inch from folded edge. There shall be no space between flaps.	301	LSb1-2	10-12	30	30
	d. The finished dosimeter pocket flap shall measure 2-5/8 (± 1/8) inches wide and 2-1/8 (± 1/8) inches deep.					
	e. The finished film badge pocket flap shall measure 3-1/8 (± 1/8) inches wide and 2-1/8 (± 1/8) inches deep.					
6.	Front-Back					
	a. Double stitch center seams of fronts and backs with lefts lapped over rights.	301 or 401	LSc-2	10-12	30 30	30 50
	b. For 4-piece construction: Double stitch right and left fronts to right and left backs respectively with fronts lapped over backs.	301 or 401	LSc-2	10-12	30 30	30 50
*	c. Cut elastic webbing 15 inches long. Position webbing on backs to correspond to marks on pattern and stitch with 4 equidistant rows of stitching along webbing. Stitch ends of webbing 1/16 to 1/8 inch from ends.	301	SSau-4	10-12	30	30

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Table II - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
7.	Sleeves					
	a. Double stitch the top seam and bottom seam of sleeves with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 30	30 50
	b. Cut wristlet 8-1/2 inches long and double. Overedge stitch the end of sleeve to knit wristlet. The finished wristlet shall measure 4 (± 1/4) inches	503 or 504	SSa-1	12-14	50	50
8.	Armholes Double stitch the sleeves to armholes with body lapped over sleeves.	301 or 401	LSc-2	10-12	30 30	30 50
9.	Inseams Double stitch the leg inseams in one continuous operation with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 30	30 50
10.	Drawcord channel - top hem a. Secure grommet and washer approximately 1-1/2 (± 1/16) inches from top raw edge, centered on the center front seam.					
	b. Turn under 1/4 inch top edge, fold over to make 3/4 (± 1/8) inch wide hem and stitch 1/16 inch from bottom folded edge.	301	EFb-1	10-12	30	30
	c. Insert drawcord into hem, allowing for 10 inches of free end of cord and knot the ends so as not to pull out of grommet.					

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Table II - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
11.	Hanger loop a. The loop shall be made by folding a piece of the basic fabric to the center with the edges abutting and stitched with a covering stitch on the underside.	406	EFh-1	10-12	50	50
	Or					
	b. Fold hanger stripping in half with the raw edges turned in and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-12	30	30
	c. Stitch each end of the hanger loop to the inside back of coverall centered over the center back seam with 3/8 inch vertical bartacks, simultaneously catching the channeled nylon drawcord.	Bartack		28 per bartack	50	50
	d. Hanger loop shall measure finished 1/4 to 3/8 inch wide and shall have a free opening of 2 1/2 to 2-3/4 inches.					
12.	Elastic loop - leg hem a. Cut elastic webbing 11-1/2 inches long. Backstitch or bartack across the width to form continuous loops 11 inches circumference with one end of the elastic webbing overlapped 1/2 inch.	301 or Bartack	LSa-1	10-12 28 per bartack	30 50	30 50

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Table II - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
12.	Elastic loop - leg hem (cont'd) b. Made a 3/4 (± 1/8) inch hem on each leg bottom with the raw edge turned under 1/4 to 3/8 inch. Stitch 1/16 inch from inside folded edge simultaneously inserting the formed elastic loop without catching it in the stitching.	301	EFb-1	10-12	30	30
13.	Foot stirrup straps Cut elastic webbing 10 1/2 (± 1/4) inches long. Position ends on inside leg hem inseam and on opposite side of leg. Stitch, 1/4 inch from raw edge, two 3/8 inch long bartacks at each end of strap along hem seamline.	Bartack		28 per bartack	50	50
14.	Hood anchor point Position 2 by 4 inch "pile" tape fastener to center back of coveralls as indicated by marks on patterns. Stitch 1/16 to 1/8 inch from edge all four sides of tape.	301	LSbj-1	10-12	30	30
15.	Label Position the top edge of the combination label on the inside of the coverall adjacent to the folded edge of drawcord channel hem and centered over the back joining seam. Stitch the label on all four sides 1/16 inch from edge. Top edge of the label may be caught when performing operation 10.c.	301	LSbj-1	10-12	30	30

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Table II - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
16.	Radiation symbol Stamp radiation symbol, as indicated in Figure 2 on the upper left back section of coverall midway between the center back and sleeve-armhole seams with the bottom of symbol aligned with the base of armhole.					
17.	Cleaning a. Trim all thread ends. b. Remove all loose threads. c. Remove all spots and stains.					

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3.8 Sizes and measurements. Unless otherwise specified, finished measurements of coveralls shall be as listed in Table III. All measurements and tolerances are expressed in inches.

Table III - Sizes and measurements

	Small-Medium	Large - X-Large	Tolerance
(A) Chest	52	56	± 1
(B) Seat	53	57	± 1
(C) Leg inseam	31	33	± 1/2
(D) Sleeve inseam	25	25	± 1/2
(E) Back length	64-1/2	70	± 1/2

NOTE: Refer to Figure 1 for A, B, C, D, and E measurements. The above measurements shall be taken with the coverall laid out smooth and flat and the neck opening fully extended.

(A) Chest - Twice the measurement taken across front at base of armholes from folded edge to folded edge.

(B) Seat - Twice the measurement taken across front six inches above the crotch seam from folded edge to folded edge.

(C) Leg inseam - Measurement shall be taken from junction of seams (crotch) along inseam to bottom of hemmed coverall legs.

(D) Sleeve inseam - Measurement shall be taken from the base of the armhole along inseam to bottom folded edge of knitted cuff.

(E) Back length - Total back length from bottom of leg hem to top of hem at neck opening - four inches from center seam.

* 3.9 Workmanship. The finished coveralls shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirement specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection The first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance and dimensional requirements.

*4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

*4.4.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. The contractor shall furnish a certificate of compliance for the physical requirement of the nylon braid specified in 3.3.5.

*4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.3. The sample shall be one finished pair of coveralls. The lot size shall be expressed in units of coveralls.

*4.4.2.1 Visual examination The coveralls shall be examined for defects in color, design, material, construction, workmanship, and marking and the defects classified in accordance with the list below:

Defect	Classification	
	Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
a. Any weakening material defect, such as a smash, hole, cut, tear, burn, loose slub, mend, thin place, needle chew or drill hole	X	
b. Any outside part shaded		X
c. One or more broken or missing yarns	X	
d. Any nonweakening material defect such as knot, misweave, unsightly slub, loose yarn, coarse yarn, area of no dye penetration, dye streak or woven in waste		X

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Defect	Classification Major	Minor
CLEANNESS		
a. Any spot or stain on outside		X
b. Thread ends not trimmed to 1/4 inch or loose threads not removed (to be scored only when condition exists on major portion of garment)		X
c. Two or more shade or size tickets not removed		X
d. Shade marking exposed and visible on outside of coverall		X
CUTTING		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X	
COMPONENT AND ASSEMBLY		
a. Any component part omitted or required operation omitted (unless otherwise classified herein)	X	
b. Any component part twisted, distorted, pleated, tight or full		X
c. Any component not as specified (type, size, color, etc)	X	
d. Any operation not as specified (unless otherwise classified herein)		X
SEAMS AND STITCHING		
a. Accuracy of seaming		
1. Seam puckered, irregular, wavy or pleated (unless otherwise classified herein)		X
2. Part of coverall caught in any unrelated stitching operation	X	
3. Permanent fold or pleat in cloth (caught in seaming or stitching)		X
4. End of stitching when not caught in another seam or stitching backstitched less than 3/8 inch		X
5. Ends of a continuous line of stitching overlapped less than 1/2 inch		X
6. Thread used on outside not specified color		X
7. Needle perforations visible on outside		X

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Defect	Classification Major	Minor
SEAMS AND STITCHING (cont'd)		
b. Gage of stitching and seam allowance		
1. Gage of stitching irregular, not uniform or not within range specified by 1/16 inch or more (score when condition occurs on more than half the length of seam)		X
2. Edge of top stitching sewn too close to edge, resulting in damage to cloth		X
3. Width of seam allowance not as specified by 1/16 inch or more (score when condition occurs on more than half the length of seam)		X
c. Open seam 1/		
1. 3/4 inch or less		X
2. More than 3/4 inch	X	
d. Runoff, skipped or broken stitches		
1. On joining seam score as open seam 1/	-	
2. Edge or top stitching (not resulting in an open seam)		X
e. Raw edges		
1. On outside:		
a. -3/4 inch or less		X
b. -more than 3/4 inch	X	
2. On inside:		
-more than 1 inch		X
f. Seam and stitch type		
1. Not specified seam or stitch type		X
2. Looper thread of 401 stitch type finishing on the outside of the coverall		X

1/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur. On double stitched seams a seam is considered open when one or both stitches of the seam is opened.

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Defect	Classification Major	Minor
SEAMS AND STITCHING (cont'd)		
g. Stitch tension		
1. Loose tension resulting in a loose seam	X	
2. Loose tension (exposed loops of top or bottom threads)		X
3. Tight tension (stitches break when normal strain is applied in direction of the seam or stitching)	X	
h. Stitches per inch (score only when condition exists on more than half the length of the seam)		
1. Less than minimum specified		X
2. Any number in excess of the maximum specified		X
i. Direction of seam lap		
Any seam not lapped in the direction specified		X
j. Crotch seam		
Crotch seam and seat seam staggered more than 1/2 inch (measured from center to center of seams)		X
ELASTIC LEG LOOP AND HEM		
a. Loop not continuous	X	
b. Loop caught in hem stitching	X	
c. Loop less than 11 inches in circumference		X
d. Hem less than 5/8 or more than 7/8 inch in width		X
ELASTIC WEBBING (BACK)		
a. Other than 4 rows of stitching	X	
b. Not stitched under tension to 24 inch length	X	
c. Other than specified width or length		X
FOOT STIRRUP STRAPS		
a. Bartack missing, insecure, misplaced not serving intended purpose	X	
b. Misplaced by more than 1/4 inch	X	
c. Other than specified width or length		X

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Defect	Classification Major	Minor
FASTENER TAPE		
a. Misplaced by more than 1/4 inch	X	
b. Foreign matter (thread, etc.) trapped on hook tape		X
c. Other than type, length or width specified	X	
d. Other than color required for specified size		X
GROMMET		
a. Broken, or not securely clinched to fabric	X	
b. Any sharp or rough edge	X	
c. Too tightly clinched cutting fabric	X	
d. Not on center front seam		X
e. Does not provide clear access into hem	X	
HANGER LOOP		
a. Off center with center of back seam by 3/4 inch		X
b. Free opening under 2 1/2 inches or over 2 3/4 inches		X
c. Width of hanger loop under 1/4 inch or over 3/8 inch		X
DRAWCORD		
a. Omitted	X	
b. Not caught by loop bartacks	X	
DRAWCORD CHANNEL		
a. Channel hem less than 5/8 or more than 7/8 inch wide		X
b. Cord ends not knotted to form a continuous loop		X
c. Free end of drawcord less than 10 inches in length with neck opening fully extended		X
d. Drawcord caught in hem stitching	X	
POCKETS		
a. Not positioned properly by 1/2 inch or more		X
b. Dosimeter pocket (inside measurement):		
1. Less than 4 1/2 inches deep	X	
2. More than 5 inches deep		X
3. Less than 1 7/8 inches wide	X	
4. More than 2 1/8 inches wide		X

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Defect	Classification Major	Minor
POCKETS (cont'd)		
c. Film badge pocket (inside measurement)		
1. Less than 3 1/4 inches deep	X	
2. More than 3 3/4 inches deep		X
3. Less than 2 1/2 inches wide	X	
4. More than 2 3/4 inches wide		X
POCKET FLAPS (dosimeter and film badge)		
a. Two flaps not abutted or out of alignment		X
b. One or both flaps less than 2 inches or more than 2 1/4 inches deep		X
c. Dosimeter flap less than 2 1/2 inches or more than 2 3/4 inches wide		X
d. Film badge flap less than 3 inches or more than 3 1/4 inches wide		X
SLEEVES		
Sleeve pleated or puckered at armhole		X
KNITTED WRISTLET		
a. Not type specified		X
b. Less than 3 3/4 inches or more than 4 1/4 inches in length		X
c. Not double thickness	X	

* 4.4.2.2 Size measurement defects

a. Any measurement deviating from dimensions and tolerances specified in Table III shall be scored as a size measurement defect.

b. Sleeve and leg inseam lengths uneven by 1/2 inch or more shall be scored as a size measurement defect.

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*4.4.3 Acceptable quality levels and inspection levels The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units (DHU) shall be as follows:

	Inspection level	AQL
For visual examination in 4.4.2.1:		
Major	II	2.5
Total (Major and Minor combined)	II	15.0
For measurement examination in 4.4.2.2:		
One class defect	S-3	4.0

*4.4.4 Examination of preparation for delivery requirements An examination shall be made to determine that packaging, packing and marking complies with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units (DHU).

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling or polyethylene bags improperly sealed. Bulging or distorted containers.
Content	Number of coveralls per shipping container is more or less than required. Size shown on one or more coveralls not as specified on shipping container. 1/

1/ For these defects, two shipping containers in the sample shall be examined.

5. PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A Each coverall shall be folded as follows: (1) Lay coverall flat, face down. (2) Fold sleeves over coverall so that outseams meet at the center seam or back. (3) The outer edges of legs and body shall be adjusted by folding so that width of coverall measures approximately 14 1/2 inches. (4) Fold coverall in half by bringing legs up to the top. (5) Fold coverall in thirds by bringing the bottom up and the top down with the combination label visible for ready identification. The completely folded coverall shall measure approximately 11 1/2 by 14 1/2 inches. (6) Each folded pair of coveralls shall be individually packaged in a clear, snug-fitting polyethylene film bag of 0.0015 (± 20 percent tolerance) inch thickness. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. As an alternative, the polyethylene bag may be of the tuck-in or reverse flap type, where heat seal closure and corner hole are not required.

5.1.2 Level C Coveralls shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A Twenty coveralls, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed, and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The liner shall be type CF, class weather-resistant, in accordance wtih the requirements specified in MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and ten in depth within the shipping container. The contents of each container shall be completely covered on top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same shipping container.

* 5.2.2 Level B Twenty coveralls, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The liner shall be type CF, class domestic in accordance with the requirements specified in MIL-B-17757. Level A packages shall be packed flat, two in length, one in width and ten in depth within the shipping container. The contents of each container shall be completely covered on top and bottom with a sheet of commercial grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

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5.2.3 Level C (Commercial packing) Coveralls, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

* 5.3.1 Polyethylene bagged packages The polyethlene bagged packages, specified in 5.1.1, shall have the required information legibly printed or stamped in black bold letters, 1/4 inch in height, directly on the bag across the center face or on a white paper label inserted within the bag to permit ready identification. The bag or label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
SIZE AND LENGTH
QUANTITY

5.3.2 Labels, mixed sizes Each shipping container, packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stencilling, a white paper label 5 by 4 inches, with the words "MIXED SIZES" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use The coveralls covered by this specification are intended to be used as a protective garment by personnel working in low level radioactive contaminated areas in conjunction with the hood, gloves, and footwear items of the related radioactive contaminants protective outfit.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes required (see 1.2).
c. When first article is required (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples and patterns For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

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*6.4 Recycled material It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

*6.5 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irresepective of the marginal notations and relationship to the last previous issue.

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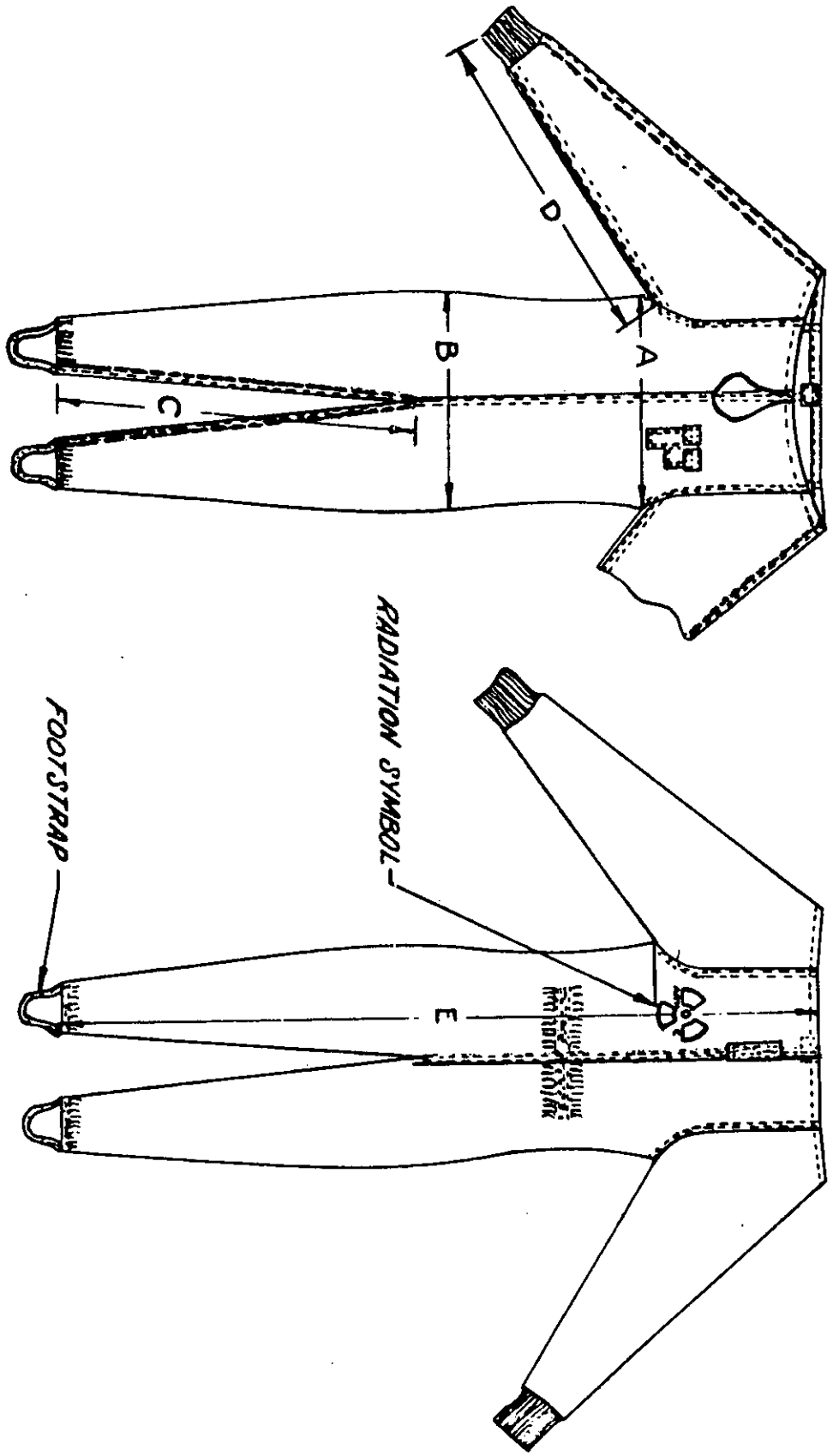


FIGURE 1. COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE

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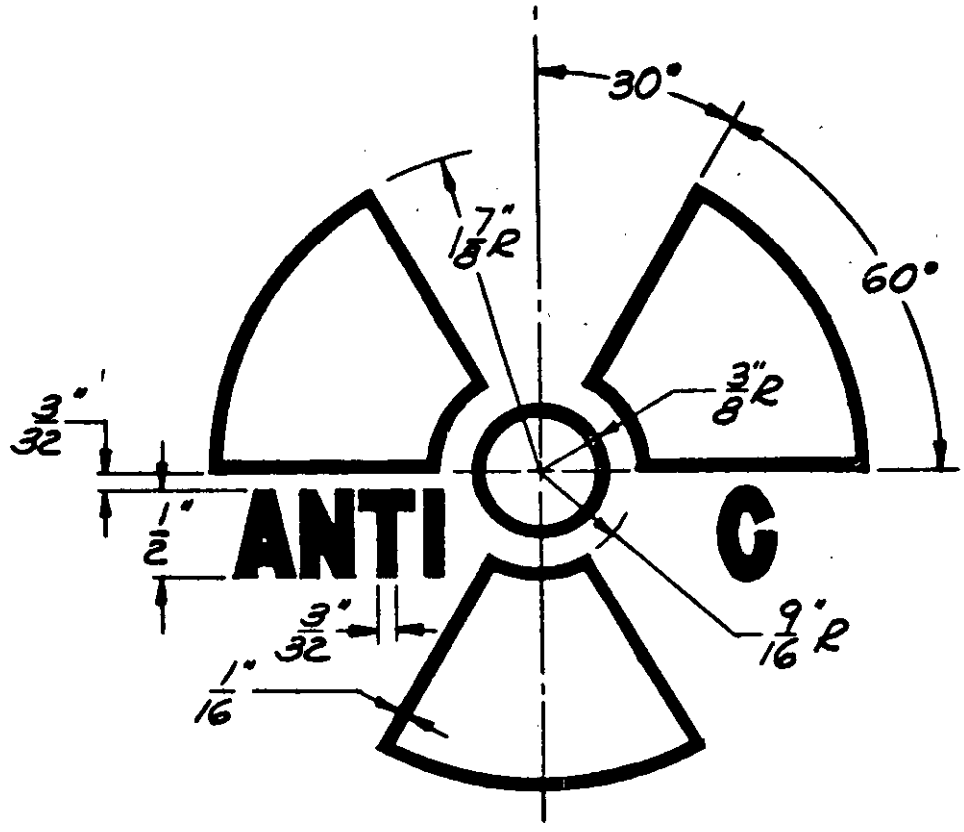


FIG. 2 RADIATION SYMBOL

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